

# Investigating Intersection-Based Hybrid Selection for Mixed Reality

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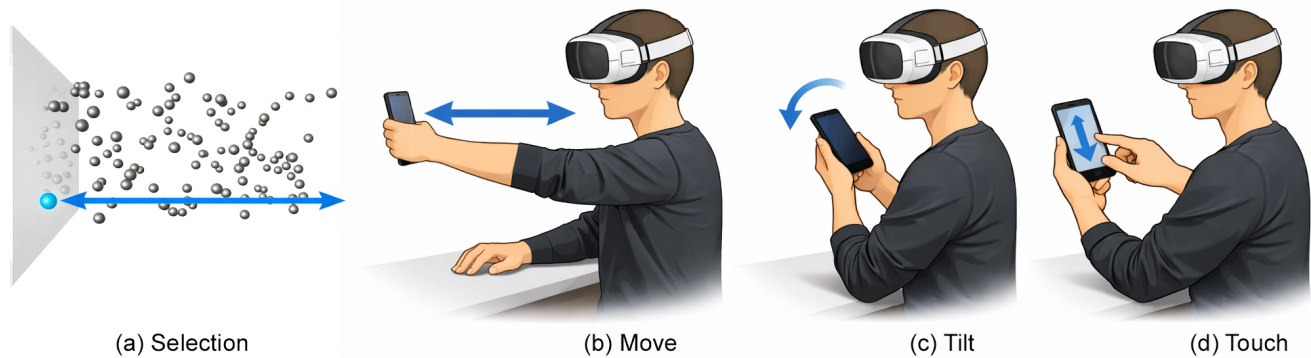
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**Figure 1:** To support ergonomic and accurate 3D selection in a cluttered space, a movable 2D plane (a) presents a vertical “slice” of objects on the smartphone screen, enabling target selection by touch. To control the plane’s distance from the user, we consider three previously explored smartphone-based techniques: (b) physically *moving* the smartphone, (c) *tilting* the smartphone, and (d) panning *touch* gestures.

## Abstract

3D selection in head-worn mixed reality environments remains challenging, as interaction techniques must carefully address ergonomics, selection performance, and object occlusion. Recent work shows that a two-dimensional selection plane can serve as an effective intermediary, reducing 3D selection to a constrained 2D interaction space. However, existing approaches have primarily explored this in the context of gaze- and hand-based input, which are prone to tracking inaccuracies. We investigate how 3D selection via a 2D plane can instead leverage established touch-based selection and smartphone panning techniques in a hybrid user interface. Taking advantage of the familiarity and precision of touch-based interaction, we argue this hybrid approach could represent a viable alternative for 3D selection. To this end, we discuss three techniques

for controlling the movement of the 2D selection plane within 3D space, and, informed by a feasibility study with 18 participants, we identify interaction conflicts that lay the groundwork for future empirical evaluation of their effectiveness.

## CCS Concepts

• Human-centered computing → Mixed / augmented reality.

## Keywords

Mixed Reality, Hybrid User Interface, 3D Selection

## ACM Reference Format:

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## 1 Introduction

Selecting objects is a fundamental task in mixed reality systems [20]. Especially in domains such as immersive analytics, users often



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need to precisely select individual items within dense, spatially distributed data. However, 3D selection through spatial interaction (e.g., controllers [34, 41] or gaze [33]) becomes increasingly difficult as the distance between users and target objects increases, as input imprecision, visual occlusion, and ergonomic constraints are amplified, thereby reducing both accuracy and usability.

To address these challenges, recent work proposed using a two-dimensional plane as an intermediary for 3D selection [2, 33, 41]. By moving a 2D plane through the 3D scene, users can reduce the complexity of the selection task to choosing targets on a planar surface, while the third axis, defining the depth or distance of the selection, is specified by the plane's position itself. Yet, existing approaches have primarily explored this interaction using bimanual mid-air controllers [41], or free-hand input combined with gaze [33]. These techniques reportedly suffer from limitations related to fatigue, precision, and sustained ergonomic comfort, particularly during prolonged or fine-grained interactions. For example, while gaze-based interaction can offer ergonomic advantages over controller-based input, it quickly becomes error-prone in densely occluded environments [33]. Tangible interaction, by contrast, has demonstrated advantages for tasks requiring precise and detailed manipulation [2], suggesting that physical input modalities may better address these challenges.

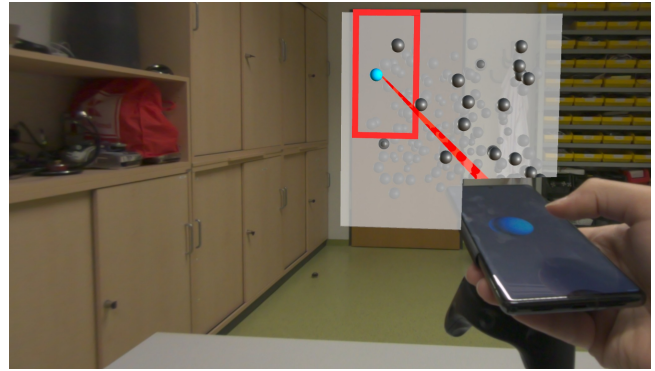
In our work, we investigate how we can leverage a hybrid user interface [12, 17] to complement the head-mounted display with a smartphone that serves as both a controller for positioning the intermediary 2D plane and a precise input device for selecting targets on that plane. By decomposing the task into (1) accurate 2D selection via touch-interaction and (2) linear movement of the plane within the 3D volume (see Fig. 1), we can build and extend well-established interaction techniques. We argue that this combination has potential for accurate and ergonomic 3D selection, but realizing this potential depends on resolving the interaction conflicts that arise when familiar touch input is combined with spatial control of the plane. To begin exploring this design space, we discuss three techniques for controlling the plane's depth as proposed by prior work, which we evaluate in a feasibility study with 18 participants. Our study reveals interaction conflicts between plane movement and target selection as a central concern for future empirical comparison.

## 2 Hybrid Pointing Techniques

We employ a planar surface [41] to decompose the task of 3D selection into two subtasks: (1) defining the selection depth by moving the 2D plane through 3D space, then (2) selecting the target on a 2D plane. While both tasks are well-established in prior work, the combination introduces additional challenges that require empirical investigation.

### 2.1 Moving the 2D Plane

To define the depth (i.e., distance from the user) of the intermediary 2D plane in 3D space, we discuss three techniques for linear relative movement [29] based on prior work [10]: *move*, *tilt*, and *touch* (see Fig. 1). Beyond this translation along the depth axis, the horizontal and vertical position of the slice is defined by raycasting a peephole (i.e., pointing the phone towards the intermediary plane,



**Figure 2: The smartphone displays a mirrored portion of the 2D plane, enabling object selection via touch input through a *peephole*. The peephole (i.e., the section of the plane that is currently mirrored onto the smartphone, outlined in red) is positioned via raycasting based on the smartphone's pointing direction (red line). This is tracked using a spatial tracker attached to the smartphone with a custom case.**

see Sec. 2.2). To avoid interference between peephole positioning and plane manipulation in the *move* and *tilt* techniques, which involve motion of the device, we implemented a *clutching* technique: our plane translation techniques are active only while a finger is touching the smartphone screen.

**2.1.1 Move.** While clutching, users can move the smartphone forward and backward within their arms' reach to translate the plane along the depth axis. Depending on the extent of the 3D volume, this mapping can be adjusted using a control–display ratio other than 1:1, to reduce the need for repeated clutching. Similar techniques have been explored in prior work on remote cursor control [1] and panning [10], or selection [5, 28] in three-dimensional spaces, and are also commonly found in tangible interaction paradigms [36].

**2.1.2 Tilt.** With this technique, users move the plane forward by tilting the smartphone down toward a more horizontal orientation, and backward by tilting it up toward a vertical orientation. To address the limited range of rotational hand motion, a control–display ratio must be applied to effectively map phone tilting to plane movement. Still, multiple clutching operations may be necessary to achieve a balance between accuracy and speed. Compared to the *move* technique, which can require substantial physical arm movement, tilting offers a more ergonomic alternative; however, it may occasionally interfere with peephole positioning when users disengage the clutch without returning the phone to its prior orientation. This approach builds on various prior work, such as demonstrating effective use of tilting to control a cursor on a distant 2D display [8], as well as in 3D space [11, 25, 32].

**2.1.3 Touch.** The *touch* technique allows users to move the intermediary plane with a single-finger vertical swipe. While this approach can offer advantages in terms of accuracy, ease of use, and physical effort, it may conflict with object selection on the smartphone, as both tasks rely on touch input. This can be partially

addressed by setting appropriate thresholds for touch gesture detection (e.g., selecting with short tap, moving with long hold and drag). Panning gestures are well-established in touch interaction and have also been successfully employed in a range of hybrid user interfaces [10, 18, 19].

## 2.2 Selection on 2D Plane

While smartphones have also been employed directly for 3D selection, their precision degrades in dense, highly occluded scenes [24, 26, 38]. To address this, 2D planar surfaces have been introduced as peepholes into the 3D scene, reducing selection to a familiar 2D task [31]. Prior work has explored the use of mobile devices for selecting items on distant surfaces or displays in a variety of contexts, for example, by mirroring digital objects onto the mobile device [9, 18] or by controlling a remote cursor [32, 37]. In particular, research on cross-device interaction has demonstrated the effectiveness of smartphones for interacting with distant 2D displays [1, 7, 8, 14, 27]. Building on these approaches, we employ an intermediary 2D plane that defines a vertical slice of the 3D target space, thereby appearing as a display facing the user [22]. We use a raycasting-based technique to position a peephole over this plane: by pointing the smartphone towards the plane, a section of the slice is mirrored onto the smartphone display [7, 14]. This enables users to precisely select targets on the screen via touch input (see Fig. 2), while holding the device in a comfortable, neutral posture, and also benefits from the passive haptic feedback of the touchscreen: unlike mid-air gaze- or hand-based selection, which requires holding the pointing limb steady in free space, the touchscreen surface provides a stable physical support against which the finger can rest and brace, supporting more controlled and precise touch input [2]. However, the dual role of touch input (i.e., both for clutching plane movement and confirming target selection) may be a source of accidental selections. We thus first conduct an initial feasibility study to investigate this overlap, before we can move on to more extended laboratory experiments.

## 3 Feasibility Study

Given many uncertainties regarding the design parameters of our techniques and their interactions with peephole navigation, we conducted a feasibility study to gain initial insights. Our study conditions correspond to the three plane-movement techniques described above. To this end, we assessed participants' perceived physical exertion (via Borg CR10 questionnaire [6]), as well as overall user experience, and qualitative aspects of using the smartphone as a secondary display via a concluding semi-structured interview. We selected the Borg CR10 [6] primarily for its low overhead between conditions, making it a suitable instrument for an early feasibility study, allowing users to rate their physical exhaustion on a category-ratio scale from 0 (*nothing at all*) to 10 (*extremely strong / maximal*). While we initially aimed to also analyze objective performance metrics (e.g., selection time, error rate, accuracy), our collected data revealed that the unresolved overlap between touch input for clutching and target selection produced a significant rate of accidental selections in our current prototype: the same touch that participants used to clutch and drag the plane could be misregistered as a target confirmation, so that a target was

selected the moment the finger was lifted or shifted rather than when intended. Since this could systematically inflate error rates and distort selection times, any objective metrics collected with the current prototype would misrepresent the techniques' potential. We therefore first concentrate on formative insights regarding the overall design of the conditions, thus laying the groundwork for a more exhaustive empirical comparison of our conditions in future work.

## 3.1 Participants

We recruited 18 participants (8 female, 10 male) aged 20–32 years ( $M = 25.17$ ,  $SD = 3.83$ ) from the local university, including students from computer science, biology, and economics. All participants reported normal or corrected-to-normal vision. Fifteen participants were right-handed, two were left-handed, and one was ambidextrous. All participants used a smartphone in their daily lives, rating their familiarity at  $M = 4.61$  ( $SD = 0.62$ ) on a 5-point Likert scale. Twelve participants (67%) reported prior experience with AR applications, rating their experience at  $M = 3.15$  ( $SD = 1.10$ ) on a 5-point Likert scale.

## 3.2 Task

Participants were tasked with selecting a highlighted target among a set of distractors within a 3D volume. To complete a selection, they first positioned the intermediary 2D plane along the depth axis, using one of the three movement techniques described above. Once the target was within the slice, participants positioned the peephole so that it appeared mirrored on the smartphone screen, where they could tap to select. To distinguish intentional selections from clutching interactions, we employed a movement-based threshold.

## 3.3 Apparatus and Dataset

We used a Varjo XR-4 HMD connected to a desktop computer, and a Google Pixel 7 Pro smartphone (6.7 inch). For tracking the smartphone position in 3D space, it was placed in a custom-built case with an HTC Vive Tracker 2.0 attached (cf. [35], see Fig. 2). We manually calibrated the position and orientation in software prior to the study.

In line with prior work [33, 41], our study prototype, implemented in Unity with Colibri for networking [16], includes a tutorial task set (10 points with no distractors) and six study task sets: two per condition, investigating two point densities (75 and 150 points) [4, 21, 30, 39]. Each task set consists of 20 selection tasks, resulting in 120 selections per participant ( $3 \times 2 \times 20$ ). Point positions were generated semi-randomly, ensuring a minimum distance of 13 cm between targets and distractors to avoid overlap. The target was highlighted in blue among gray distractors, which all measured 8 cm in diameter. We fully counterbalanced conditions, but kept the task set order fixed. We ensured comparability within and between task sets with a constant ideal movement distance across sets. The full 3D volume measured 1.35 m in width and height, had a depth of 4 m, and was placed approximately 1.5 m in front of participants. The peephole size was set to a 1:5 ratio (smartphone size to 3D volume), resulting in a peephole size of  $75 \times 34$  cm within the 3D volume.



**Figure 3: Boxplots of the Borg CR10 questionnaire. \* marks statistically significant differences and lines indicate medians.**

### 3.4 Procedure

Each session began with a brief introduction and participants giving their informed consent, and demographic information. Participants were then equipped with the HMD and smartphone and completed the tutorial training task. Then, participants completed 20 tasks in the low-density task set for each condition, followed by 20 tasks in the high-density task set for each condition. They were prompted to perform the selections as quickly as possible. After completing each condition, participants removed the HMD and filled out the Borg CR10 questionnaire. Participants were seated at a table throughout the session, allowing them to use it as an armrest. We concluded the session with a semi-structured interview to gather qualitative feedback in four themes:

- Perceived workload across techniques (e.g., “How exhausted did you feel from holding the smartphone?” and “Which of the techniques did you find the most exhausting?”);
- Subjective preference and ranking (e.g., “In which order would you rank the three selection techniques?”);
- The combined use of HMD and smartphone (e.g., “How well could you handle using two devices at the same time?” and “Did the phone screen bring you added value?”); and
- Visual attention switching between the two devices (e.g., “How often and in which cases did you look at the phone, and did this differ between techniques?”).

We captured responses as written notes by the experimenter and subsequently grouped by participant and theme for analysis. All procedures were conducted in accordance with the ethical and sanitary guidelines of the local university. Each session took approximately 90 minutes and participants were offered monetary compensation for their time.

### 3.5 Results

We report on our results based on our focus on *perceived workload*, overall *user experience*, and the *role of the smartphone*.

**3.5.1 Perceived Workload.** Overall, participants reported a relatively low perceived workload for the *touch* and *move* techniques, whereas *tilt* was frequently perceived as more physically and mentally exhausting. Even though *move* requires more physical arm movement, eight participants (44%) specifically reported that *tilt* was particularly tiring. Further, two participants mentioned feeling more confident using *move* than *tilt*. These perceptions are also partially reflected in the Borg CR10 ratings (see Fig. 3): Here, we confirmed normal data distribution using a Shapiro-Wilk test and conducted a repeated-measures ANOVA, which revealed a significant effect of condition on perceived physical exertion ( $F(2) = 13.282, p < .001$ ). Subsequent post hoc analysis with a pairwise t-test

(assuming an alpha value of 0.017 due to Bonferroni correction) showed significantly lower exertion with *touch* compared to both *tilt* ( $t(17) = -5.23, p < .001$ ) and *move* ( $t(17) = -3.30, p = .013$ ). This pattern is consistent with prior reports that mid-air spatial input typically incurs higher physical demand than touch [8].

**3.5.2 User Experience.** Participants appreciated *touch* for its familiarity, and liked *move* as their physical hand motion directly mirrored the movement of the intermediary plane. In contrast, *tilt* was viewed more critically: Eight participants (44%, four of which also noted that *tilt* was particularly tiring) reported confusion regarding the correspondence between the degree of phone rotation and the resulting plane movement, as well as the mapping between tilting orientation and translation direction. In addition, two participants found *tilt* insufficiently accurate, while another two found it overly sensitive.

Beyond overall preference, several participants reported issues related to clutching, in particular as it led to several accidental selections. Although we did not assess accuracy or performance quantitatively in this pilot study, nine participants (50%) reported making selection errors specifically due to the interference of touch gestures for both clutching and target selection. For example, one participant described accidentally confirming a target on releasing the clutch: “sometimes I would accidentally press one of the gray spheres which I’m not supposed to press [...] that [selection] is accidental because I’m just letting go after scrolling”. Another participant emphasized how often this occurred and attributed it to the split of attention between the two displays: “[the accidental selection] happens quite often for me [...] because I only want to move the plane, and I cannot look at both the AR-view and this phone view at the same time”. This issue was frequently observed in situations with multiple distractors on the plane, suggesting it may be especially problematic in dense environments.

**3.5.3 Use of Smartphone.** Feedback on the use of the smartphone as part of the hybrid interface was mixed. Ten participants (56%) appreciated this combination of complementary devices, with one comparing the smartphone to a television remote. In contrast, four participants did not perceive the smartphone as useful; five reported rarely using it during the tasks, and three indicated that they only looked at the phone at the beginning of the study.

Participants also raised concerns regarding visual attention switching: Nine participants (50%) reported looking at the smartphone when selecting targets, while four stated that they only consulted the screen when many distractors were present nearby. One participant noted that shifting gaze to the phone disrupted their interaction flow, and another described extending their arms to keep both the 3D volume and the smartphone display within their field of view.

## 4 Lessons Learned and Future Work

In the following, we summarize key lessons from our findings and outline directions for future empirical evaluation.

### 4.1 Peephole Navigation

Although we assume that using a smartphone as a hybrid user interface [17] can offer increased precision, it also introduces an

additional screen that demands the user’s visual attention during selection. This can lead to ergonomic challenges: For example, to position the peephole via raycasting, some participants used the smartphone like a mid-air controller, keeping both the device’s display and the 3D space within their field of view. Such a posture can substantially increase physical demand on the arms and may lead to fatigue commonly described as the “gorilla arm syndrome” [8]. Alternatively, users may hold the smartphone in a more comfortable, low position near the hip. While this likely reduces arm strain, it may increase neck strain and the need for visual switching, so that ergonomic load is merely shifted rather than eliminated, and mental load is potentially increased.

Thus, we see the need to re-evaluate navigation through raycasting within the context of hybrid user interfaces and investigate potential mitigation mechanisms, such as intuitive mode switching [13] or eyes-free interaction techniques [18]. For example, providing preview cues on the intermediary plane (e.g., similar to the peephole outline) before the user’s finger touches the touchscreen [15] has been shown to reduce the need for sustained visual attention to the device, thereby alleviating some of the associated ergonomic burden [23]. Alternatively, other peephole navigation techniques, such as touch-based panning and zooming [19], could be explored to avoid overlap with the investigated plane movement methods, offering a complementary approach [40] to the *move* technique.

## 4.2 Plane Movement

While our techniques for moving the intermediary plane build on prior work for 3D object manipulation, each introduces distinct trade-offs, particularly when combined with peephole navigation. Specifically, both the *tilt* and *move* techniques have different overlaps with the raycasting-based peephole navigation, whereas *touch* interferes with target selection. Although this can be compensated by design strategies, such as clutching or thresholding, our findings show that this can quickly make selection more error-prone (e.g., accidental selections) or reduce perceived performance (e.g., as *tilt* resets the raycasting position). Based on initial feedback from our study, we argue that *move* could serve as a suitable complementary approach [40] to peephole navigation using raycasting, even though this requires the most arm movement. This suggests a potential trade-off between ergonomics, performance, and precision, which warrants further investigation [3]. The *tilt* technique offers a potential balance between physical movement and ergonomics, but participants found it imprecise due to various difficulties regarding the unintuitive mapping of phone rotation to plane translation direction, or interference with phone movement for peephole navigation.

Finally, our current investigation is limited to linear translation of the plane along the depth axis. Prior work has demonstrated the benefits of more expressive plane manipulation in 3D space, including rotation or free movement [33]. Within our hybrid user interface, such degrees of freedom could be mapped to the smartphone’s rotation (e.g., similar to [2]). However, this could again interfere substantially with peephole navigation and might necessitate alternative selection techniques.

## 5 Conclusion

We investigate using a smartphone as a complementary interaction device for 3D selection in head-mounted mixed reality environments. By decomposing distant 3D selection into (1) controlled movement of an intermediary 2D plane within the 3D space and (2) accurate 2D selection on this screen-projected plane, our approach builds on well-established interaction techniques. We argue that this hybrid approach has the potential to support more ergonomic 3D selection than existing controller- and gaze-based methods, but that realizing this potential first requires resolving the interaction conflicts surfaced by our feasibility study.

To complement raycasting-based peephole navigation on the intermediary plane, we introduce three techniques for controlling the plane’s depth drawing on prior work: moving the smartphone, tilting it, or using touch-based panning. Findings from a feasibility study indicate that movement and touch-based techniques could serve as a suitable complementary approach, while tilting was least preferred. However, our findings also reveal issues due to overlapping input mapping for peephole navigation and plane movement that warrant further investigation of compensation mechanisms. We thus distill our insights into lessons learned that inform the design and evaluation of future plane-based selection techniques using hybrid user interfaces.

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